

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

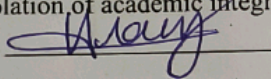
Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Mohammed Nasser ¹⁹²³²⁵⁰⁶⁰ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Cloud Architecture for Video Streaming

Architecture:-

- * Use cloud servers in multiple regions.
- * Store videos in cloud storage.
- * Use CDN for faster video delivery.

Load Balancing & Auto scaling.

- * Load balancer distributes user traffic.
- * Auto scaling adds or removes servers automatically.
- * Ensures smooth streaming during peak traffic.

Cost optimization:-

- * Use Pay-as-you-go model.
- * Archive old videos.
- * Remove unused resources.

(2) Multi-cloud container Deployment:-

Solution:-

- * Use Docker containers.
- * Manage them with Kubernetes.

* Deploy across multiple cloud providers.

Service Discovery & Scaling:-

* Kubernetes finds services automatically

* Scale pods based on workload.

Benefits & Risks:-

Benefits:-

* High availability.

* Better scalability

Risks:-

* complex management.

* Higher costs.

Q3) Backup and Disaster recovery:-

item

Backup strategy:-

* Enable automatic backups.

* Use data replications.

* Enable versioning.

RTO & RPO:-

- * RTO reduces recovery time.
- * RPO minimizes data loss.

Business continuity:-

- * Test backups regularly.
- * Monitor recovery systems.
- * Maintain recovery plans.

Conclusion:- cloud-native backup, replication and versioning ensure fast recovery with minimal downtime and data loss.

Q4) Hybrid cloud Architecture:-

Architecture:-

- * Store sensitive data on-premises.
- * Use public cloud for other applications.
- * Connect both using secure VPN.

Security:-

- * Use IAM and MFA
- * Encrypt stored and transmitted data.

Challenges:-

- * complex management
- * compliance issue.
- * Network latency

Conclusion:-

A hybrid cloud combines and scalability while meeting compliance requirements for sensitive government data.

Q5) multi-region deployment strategy for SaaS application

Deployments:-

- * Deploy application in multiple regions.
- * Use global load balancing.

Replication & failover:-

- * Replication data across regions.
- * enables automatic failover.

Monitoring:-

- * monitor servers continuously.
- * set up alerts for failures.
- * use centralized logging.

Conclusion:- A multi-region deployment ensures high availability, fault tolerance and uninterrupted service even during regional outages.